

Exact Three-Layer Reconstruction of a Semiprime

Homogeneous Layer Invariants, the Shifted Symmetric Parameter, and the Emergence
of the Factor $433^2 769^2$

Experimental Algebraic Reconstruction Study

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Abstract

We investigate an exact symbolic reconstruction experiment based on a self-contained kernel

$$G_{9,16}(N, X),$$

where

$$N = pq, \quad X = p + q + 1.$$

The experiment uses no floating-point arithmetic. For the test instance

$$p = 50387, \quad q = 282589,$$

we have

$$N = 14238811943, \quad p + q = 332976, \quad X = 332977.$$

The kernel decomposes into seventeen nonzero homogeneous layers,

$$G_{9,16}(N, X) = \sum_{d=0}^{16} L_d.$$

We show experimentally that the three highest layers

$$L_{16}, \quad L_{15}, \quad L_{14}$$

are sufficient, in this instance, to construct a scale-free invariant

$$R = \frac{L_{15}^2}{L_{16}L_{14}},$$

whose exact elimination equation has degree 18 and contains the true value

$$t = \frac{N}{X}$$

as an exact rational root.

The reconstruction yields

$$X = 332977, \quad N = 14238811943, \quad S = X - 1 = 332976,$$

and subsequently

$$\{p, q\} = \{50387, 282589\}.$$

An independently reconstructed prediction for the unused fourth layer L_{13} agrees exactly with the observed value.

A particularly notable structural observation is

$$\gcd(L_{16}, L_{15}, L_{14}, L_{13}) = 110873682529 = 332977^2.$$

Since

$$332977 = 433 \cdot 769,$$

this gives

$$110873682529 = 433^2 769^2.$$

Thus the apparently unexplained factors 433 and 769 arise from the factorization of the shifted symmetric parameter

$$X = p + q + 1,$$

rather than directly from the factorization of N .

The experiment demonstrates an exact algebraic reconstruction mechanism, but does not by itself constitute a factorization algorithm, because the layer values are supplied by direct evaluation of the kernel using the hidden semiprime configuration. The central unresolved question is whether equivalent layer information can be obtained from N alone.

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1 Introduction

Let

$$N = pq$$

be a product of two distinct primes. The classical symmetric variables associated with the factorization are

$$S = p + q$$

and

$$\Delta = (p - q)^2 = S^2 - 4N.$$

Once S is known, the factorization of N follows from the quadratic equation

$$z^2 - Sz + N = 0.$$

The present experiment studies a different representation of the same information. A symbolic kernel

$$G_{9,16}(N, X)$$

is evaluated at

$$X = S + 1 = p + q + 1.$$

The resulting expression is decomposed into homogeneous layers with respect to the relevant grading:

$$G_{9,16}(N, X) = \sum_{d=0}^{16} L_d.$$

The central experimental question is:

How much information about the hidden semiprime configuration is contained in a small number of the highest homogeneous layers?

For the test instance considered here, the answer is unexpectedly strong. The top three layers are sufficient to recover the exact rational quantity

$$t = \frac{N}{X},$$

after which X , N , S , and finally p, q can be reconstructed.

An additional structural phenomenon appears when the large integers $L_{16}, L_{15}, L_{14}, L_{13}$ are factored. Their common divisor is

$$332977^2.$$

At first sight this produces the apparently unrelated factors

$$433^2 769^2.$$

The purpose of this paper is to identify their origin and place the observation into the broader reconstruction structure.

2 Exact Symbolic Setup

We fix

$$k = 9, \quad \ell = 16.$$

The kernel is denoted

$$G_{9,16}(N, X).$$

The hidden prime pair used in the experiment is

$$p = 50387, \quad q = 282589.$$

They are distinct primes.

Their product is

$$N = pq \tag{1}$$

$$= 50387 \cdot 282589 \tag{2}$$

$$= 14238811943. \tag{3}$$

Their sum is

$$S = p + q \tag{4}$$

$$= 50387 + 282589 \tag{5}$$

$$= 332976. \tag{6}$$

The shifted symmetric parameter is therefore

$$X = S + 1 \tag{7}$$

$$= 332977. \tag{8}$$

Thus the fundamental identities for this experiment are

$$\boxed{N = 14238811943, \quad S = 332976, \quad X = 332977.} \tag{9}$$

The experiment additionally verifies the exact symbolic identity

$$F(p, q) = G(pq, p + q + 1). \tag{10}$$

The resulting value of $F(p, q)$ has 126 decimal digits.

All computations described below are performed exactly over $\mathbb{Z}$ or $\mathbb{Q}$.

3 Homogeneous Layer Decomposition

The kernel has highest degree 16 and lowest degree 0. Consequently, it decomposes into seventeen homogeneous components:

$$G_{9,16}(N, X) = \sum_{d=0}^{16} L_d. \tag{11}$$

The experiment verifies

$$L_d \neq 0 \quad \text{for every } 0 \leq d \leq 16.$$

The four highest layers are the following.

3.1 L_{16}

$$L_{16} = -229819595874131650867437335526223596013696239571803872596424049829315518497606063887810$$

3.2 L_{15}

$$L_{15} = 28858754233049457868021223714659336444940754487769148448477176799531299261964342878616877$$

3.3 L_{14}

$$L_{14} = -116239665211965343729749526675162570184786431182562624678991639860937924723605412195180$$

3.4 L_{13}

$$L_{13} = 15577154207069906934693585527875575564217205934718322058729179547613741531677187871702169$$

The signs of the four highest layers are

$$-, +, -, +.$$

4 Normalized Layer Polynomials

Introduce the dimensionless variable

$$t = \frac{N}{X}. \quad (12)$$

The top layers admit normalized polynomial descriptions in t .
For degree 16,

$$\begin{aligned} h_{16}(t) = & -9t^8 - 36t^7 - 84t^6 - 126t^5 \\ & - 126t^4 - 84t^3 - 36t^2 - 9t - 1. \end{aligned} \quad (13)$$

For degree 15,

$$\begin{aligned} h_{15}(t) = & 88t^9 + 396t^8 + 1164t^7 + 2226t^6 \\ & + 2898t^5 + 2604t^4 + 1596t^3 \\ & + 639t^2 + 151t + 16. \end{aligned} \quad (14)$$

For degree 14,

$$\begin{aligned} h_{14}(t) = & -276t^{10} - 1380t^9 - 5460t^8 - 13560t^7 \\ & - 23058t^6 - 27510t^5 - 23100t^4 \\ & - 13410t^3 - 5135t^2 - 1169t - 120. \end{aligned} \quad (15)$$

The normalized representation separates the dependence on the ratio N/X from the overall scale associated with X .

5 The Three-Layer Scale-Free Invariant

The central invariant used in the reconstruction is

$$\boxed{R = \frac{L_{15}^2}{L_{16}L_{14}}}. \quad (16)$$

The reason for this choice is that common multiplicative scale factors present in the individual layers can cancel.

Suppose schematically that

$$L_d = X^{a_d} h_d(t), \quad t = \frac{N}{X}. \quad (17)$$

Then

$$R = X^{2a_{15}-a_{16}-a_{14}} \frac{h_{15}(t)^2}{h_{16}(t)h_{14}(t)}. \quad (18)$$

For the present kernel the exponents cancel, producing a function of t alone:

$$R = R(t). \quad (19)$$

The experiment evaluates R exactly from the observed values L_{16}, L_{15}, L_{14} and independently evaluates the theoretical expression $R(t)$ at

$$t_{\text{true}} = \frac{14238811943}{332977}.$$

The two exact rational values agree.

Hence

$$\boxed{R_{\text{observed}} = R(t_{\text{true}})}. \quad (20)$$

This is the first major reconstruction result.

6 Exact Elimination Equation

Clearing denominators in the equation

$$R(t) = R_{\text{observed}} \quad (21)$$

produces an exact polynomial equation

$$P(t) = 0. \quad (22)$$

The resulting elimination polynomial has degree

$$\deg P = 18.$$

Its factorization contains the two linear factors

$$332977t - 14238811943 \quad (23)$$

and

$$332977t + 14239144920. \quad (24)$$

Thus

$$P(t) = C(332977t - 14238811943)(332977t + 14239144920)Q_{16}(t), \quad (25)$$

where $C \in \mathbb{Z} \setminus \{0\}$ and Q_{16} is a degree-16 polynomial.

The true value

$$t_{\text{true}} = \frac{14238811943}{332977} \quad (26)$$

is therefore an exact root:

$$P(t_{\text{true}}) = 0. \quad (27)$$

7 Exact Reconstruction of X and N

An exact rational-root search followed by the required consistency test produces two surviving candidates:

$$t_1 = -\frac{14239144920}{332977} \quad (28)$$

and

$$t_2 = \frac{14238811943}{332977}. \quad (29)$$

Both candidates have

$$X = 332977.$$

The corresponding values of N are

$$N_1 = -14239144920$$

and

$$N_2 = 14238811943.$$

The positive semiprime configuration is therefore

$$\boxed{t = \frac{14238811943}{332977}, \quad X = 332977, \quad N = 14238811943.} \quad (30)$$

Since

$$X = S + 1,$$

we obtain

$$\boxed{S = X - 1 = 332976.} \quad (31)$$

8 Recovery of the Prime Factors

Once N and S are known, the hidden factors are roots of

$$z^2 - Sz + N = 0. \quad (32)$$

Its discriminant is

$$\Delta = S^2 - 4N \quad (33)$$

$$= 332976^2 - 4(14238811943) \quad (34)$$

$$= 53917768804. \quad (35)$$

For a semiprime $N = pq$,

$$S^2 - 4N = (p + q)^2 - 4pq = (p - q)^2. \quad (36)$$

Therefore

$$\sqrt{\Delta} = |p - q|. \quad (37)$$

The two roots are

$$p, q = \frac{S \pm \sqrt{\Delta}}{2}. \quad (38)$$

For the present instance,

$$\boxed{\{p, q\} = \{50387, 282589\}.} \quad (39)$$

Direct verification gives

$$50387 \cdot 282589 = 14238811943, \quad (40)$$

$$50387 + 282589 = 332976. \quad (41)$$

Both recovered factors are prime.

9 Independent Prediction of L_{13}

The fourth layer L_{13} is not required for the three-layer reconstruction.

After reconstructing N and X , the kernel is reevaluated to predict L_{13} .

The predicted value is

$$L_{13}^{\text{pred}} = 1557715420706990693469358552787557556421720593471832205872917954761374153167718787170216$$

The independently observed value is exactly identical:

$$L_{13}^{\text{obs}} = 1557715420706990693469358552787557556421720593471832205872917954761374153167718787170216$$

Consequently,

$$\boxed{L_{13}^{\text{pred}} = L_{13}^{\text{obs}}} \quad (42)$$

This provides an independent validation of the reconstructed parameter pair (N, X) .

10 The Common Divisor of the Homogeneous Layers

We now turn to the observation motivating the present investigation.

The exact greatest common divisor of the four highest observed layers is

$$\gcd(L_{16}, L_{15}, L_{14}, L_{13}) = 110873682529. \quad (43)$$

Factorization gives

$$110873682529 = 433^2 769^2. \quad (44)$$

At first glance, the appearance of 433 and 769 may suggest a relationship with the hidden prime factors p and q . This interpretation is incorrect.

The key calculation is

$$433 \cdot 769 = 332977 \quad (45)$$

$$= X. \quad (46)$$

Therefore

$$\boxed{X = 433 \cdot 769.} \quad (47)$$

Squaring,

$$X^2 = 433^2 769^2. \quad (48)$$

Since

$$X = 332977,$$

we obtain

$$X^2 = 332977^2 \quad (49)$$

$$= 110873682529. \quad (50)$$

Thus

$$\boxed{\gcd(L_{16}, L_{15}, L_{14}, L_{13}) = X^2.} \quad (51)$$

This completely explains the numerical origin of the factors 433 and 769 in the common divisor.

11 Interpretation of 433 and 769

The identity

$$433 \cdot 769 = X \quad (52)$$

combined with

$$X = p + q + 1 \quad (53)$$

gives

$$\boxed{433 \cdot 769 = p + q + 1.} \quad (54)$$

Consequently,

$$433^2 769^2 = (p + q + 1)^2. \quad (55)$$

The integers 433 and 769 therefore do not originate from the factorization

$$N = pq.$$

Indeed,

$$50387 \neq 433, \quad 282589 \neq 769.$$

Instead, they arise from a secondary factorization:

$$\boxed{p + q + 1 = 433 \cdot 769.} \quad (56)$$

The correct hierarchy is therefore

$$N = pq, \quad (57)$$

while separately

$$X = p + q + 1 = 433 \cdot 769. \quad (58)$$

The observed common factor of the layers is

$$X^2. \quad (59)$$

12 Structural Hypothesis for the Layers

The numerical evidence suggests that the homogeneous layers possess an explicit X -dependent structure.

At minimum, the observed divisibility supports the hypothesis

$$L_d = X^2 Q_d(N, X) \quad (60)$$

for the layers tested.

More generally, the normalized polynomial representation suggests a form of the type

$$L_d = X^{a_d} h_d \left(\frac{N}{X} \right). \quad (61)$$

Such a representation has two important consequences.

First, the large integer magnitude of L_d contains a substantial contribution from powers of X .

Second, ratios of layers can eliminate these scale factors. The three-layer invariant

$$\frac{L_{15}^2}{L_{16} L_{14}}$$

is an example of such a cancellation.

The reconstruction therefore appears to separate naturally into two algebraic variables:

$$t = \frac{N}{X}$$

and

$$X.$$

The invariant determines the former, while the exact normalization and consistency conditions determine the latter.

13 Reconstruction Chain

The complete experimentally verified chain is

$$(L_{16}, L_{15}, L_{14}) \longrightarrow R \longrightarrow t = \frac{N}{X} \longrightarrow X \longrightarrow N \longrightarrow S \longrightarrow \Delta \longrightarrow \{p, q\}. \quad (62)$$

For the numerical instance,

$$(L_{16}, L_{15}, L_{14}) \longrightarrow \frac{14238811943}{332977}, \quad (63)$$

$$\frac{N}{X} \longrightarrow X = 332977, \quad (64)$$

$$X \longrightarrow S = 332976, \quad (65)$$

$$\frac{N}{X} X \longrightarrow N = 14238811943, \quad (66)$$

$$S^2 - 4N \longrightarrow \Delta = 53917768804, \quad (67)$$

$$\frac{S \pm \sqrt{\Delta}}{2} \longrightarrow \{50387, 282589\}. \quad (68)$$

Every step is performed exactly.

14 Experimental Verification

The experiment performs the following checks.

Check	Result
$F(p, q) = G(pq, p + q + 1)$	True
All 17 homogeneous layers nonzero	True
Observed invariant = theoretical invariant	True
$P(t_{\text{true}}) = 0$	True
X reconstructed exactly	True
N reconstructed exactly	True
S reconstructed exactly	True
L_{13} independently predicted	True
$G(N, X)$ reconstructed exactly	True
$pq = N$	True
$p + q = S$	True
$(p - q)^2 = \Delta$	True
Recovered p, q prime	True
$F(p, q)$ reproduced exactly	True

The experiment reports

$$\boxed{14 \text{ checks passed, } \quad 0 \text{ failures.}} \quad (69)$$

15 What Has Been Demonstrated

The experiment establishes the following facts for the tested instance.

- (i) The exact kernel evaluation produces seventeen nonzero homogeneous layers.
- (ii) The top three layers determine an exact scale-free invariant.
- (iii) The invariant yields an exact polynomial equation in

$$t = N/X.$$

- (iv) The true value of t is an exact rational root of the elimination polynomial.
- (v) The reconstruction recovers X exactly.
- (vi) Since $X = S + 1$, the symmetric quantity $S = p + q$ is recovered.
- (vii) The original semiprime N is recovered.
- (viii) The quadratic

$$z^2 - Sz + N$$

recovers the unordered factor pair.

- (ix) The unused layer L_{13} is reproduced exactly from the reconstructed parameters.
- (x) The common divisor of the top four layers is

$$X^2.$$

- (xi) The factors 433 and 769 arise because

$$X = 433 \cdot 769.$$

16 What Has Not Been Demonstrated

The reconstruction experiment should not yet be described as a general integer factorization algorithm.

The reason is fundamental.

The input to the reconstruction experiment includes the layer values

$$L_{16}, L_{15}, L_{14}.$$

Those layers were obtained from direct evaluation of

$$G_{9,16}(N, X)$$

using the hidden value

$$X = p + q + 1.$$

In a standard factorization problem, only

$$N$$

would be supplied.

Thus the experimentally established implication is of the form

$$(L_{16}, L_{15}, L_{14}) \implies (p, q), \quad (70)$$

not yet

$$N \implies (p, q). \quad (71)$$

The missing step is to determine whether the layer information can be computed or reconstructed efficiently from N alone.

This distinction prevents the current experiment from being interpreted as a demonstrated polynomial-time factorization method.

17 Questions for the Next Experiment

The most important next step is to remove the hidden prime pair entirely.

Given only

$$N,$$

one should investigate whether one can construct quantities equivalent to the top homogeneous layers without explicitly knowing p or q .

Several questions are particularly natural.

17.1 Can X^2 be recovered directly?

The present experiment establishes

$$\gcd(L_{16}, L_{15}, L_{14}, L_{13}) = X^2.$$

The next question is whether an analogous quantity can be derived from N alone.

17.2 Can the scale-free invariant be computed from N ?

The invariant

$$R = \frac{L_{15}^2}{L_{16}L_{14}}$$

depends only on

$$t = \frac{N}{X}$$

after cancellation of the homogeneous scale.

If an equivalent invariant can be computed without X , then the reconstruction mechanism becomes substantially more significant.

17.3 Is the X^2 factor universal?

The observed divisibility should be tested for all homogeneous layers:

$$X^2 \mid L_d \quad (0 \leq d \leq 16).$$

It should also be tested for other values of k and ℓ .

17.4 Is there a closed form for $h_d(t)$?

The normalized polynomials

$$h_{16}(t), \quad h_{15}(t), \quad h_{14}(t)$$

have highly structured integer coefficients.

A general formula for h_d could reveal the origin of the scale cancellation and perhaps explain the degree-18 elimination polynomial.

17.5 Does the elimination structure generalize?

The factorization

$$P(t) = C(332977t - 14238811943)(332977t + 14239144920)Q_{16}(t)$$

should be computed for many independent semiprimes.

One should determine whether the degree-18 structure and the two linear factors are universal consequences of the kernel.

18 Conclusion

This paper examined an exact symbolic reconstruction experiment based on the homogeneous layers of the kernel

$$G_{9,16}(N, X), \quad X = p + q + 1.$$

For the test instance

$$p = 50387, \quad q = 282589,$$

the corresponding values are

$$N = 14238811943, \quad S = 332976, \quad X = 332977.$$

The three highest layers yield a scale-free invariant that determines

$$t = \frac{N}{X}.$$

Exact elimination and rational-root reconstruction then recover X and N , after which

$$S = X - 1$$

and

$$p, q = \frac{S \pm \sqrt{S^2 - 4N}}{2}$$

recover the original prime pair.

The independent prediction of L_{13} agrees exactly with the observed value, and all fourteen reported verification checks pass.

The most direct explanation of the previously observed integers 433 and 769 is the identity

$$433 \cdot 769 = 332977 = X = p + q + 1.$$

Consequently,

$$\boxed{433^2 769^2 = X^2 = (p + q + 1)^2.}$$

Thus the factors 433 and 769 do not arise from the original factorization

$$N = pq.$$

They arise from the factorization of the shifted symmetric quantity

$$p + q + 1.$$

The deeper structural observation is therefore not the appearance of 433 and 769 themselves, but the apparent persistence of an X^2 factor in the homogeneous layers.

The central open problem is now clear:

$$\boxed{\text{Can the information encoded by } L_{16}, L_{15}, L_{14} \text{ be generated from } N \text{ alone?}} \quad (72)$$

If the answer is affirmative and the resulting quantities are efficiently computable, the reconstruction mechanism could have implications for algorithmic semiprime reconstruction. If the answer is negative, the experiment nevertheless reveals a nontrivial exact algebraic encoding of the symmetric factorization parameter $p + q + 1$ inside the homogeneous layer structure.

A Numerical Data

For completeness, the principal exact values used throughout the experiment are

$$p = 50387, \quad (73)$$

$$q = 282589, \quad (74)$$

$$N = 14238811943, \quad (75)$$

$$S = 332976, \quad (76)$$

$$X = 332977, \quad (77)$$

$$\Delta = 53917768804. \quad (78)$$

The key identities are

$$N = pq, \quad (79)$$

$$S = p + q, \quad (80)$$

$$X = S + 1, \quad (81)$$

$$\Delta = (p - q)^2, \quad (82)$$

$$X = 433 \cdot 769, \quad (83)$$

$$X^2 = 433^2 \cdot 769^2, \quad (84)$$

$$\gcd(L_{16}, L_{15}, L_{14}, L_{13}) = X^2. \quad (85)$$

The exact reconstruction is therefore

$$\boxed{\begin{aligned} L_{16}, L_{15}, L_{14} &\longrightarrow t = \frac{14238811943}{332977}, \\ t &\longrightarrow X = 332977, \\ X &\longrightarrow S = 332976, \\ (N, S) &\longrightarrow \{50387, 282589\}. \end{aligned}}$$